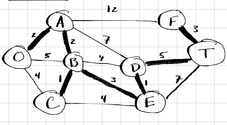
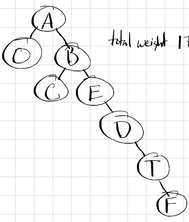
Homework 5

(Tripp) Milton Lamb CS617 Fall 24220

Problem 1

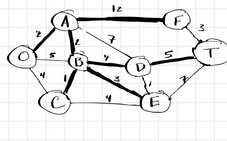


vertex	fringe
A(–, Φ)	O(A,2), B(A,2), D(A,7), F(A,12)
B(A,2)	O(A,2), C(B,1), E(B,3), D(B,4), F(A,12)
C(B,1)	O(A,2), E(B,3), D(B,4), F(A,12)
O(A,2)	E(B,3), D(B,4), F(A,12)
E(B,3)	D(E,1), F(A,12), T(E,7)
D(E,1)	F(A,12), T(D,5)
T(D,5)	F(T,3)
F(T,3)	

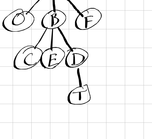


total weight 17

Problem 2



vertex	fringe
A(–, Φ)	O(A,2), B(A,2), D(A,7), F(A,12)
B(A,2)	O(A,2), C(B,3), E(B,5), D(B,6), F(A,12)
O(A,2)	C(B,3), E(B,5), D(B,6), F(A,12)
C(B,3)	E(B,5), D(B,6), F(A,12)
E(B,5)	D(B,6), F(A,12), T(E,12)
D(B,6)	F(A,12), T(D,11)
T(D,11)	F(A,12)
F(A,12)	



total distance from A - 41

Problem 3

given an undirected graph of

$$G = (V, E)$$

and an integer k

determine if a subset $S \subseteq V$ of size k exists so that no 2 $v \in S$ are adjacent

Verification in Poly Time

1. we can verify S is size k in linear time by counting the elements
2. we can verify no edge exist between vertices $u, v \in S$ by comparing all pairs $v, v \in S$ also satisfy $(u,v) \in E$ in $O(V^2)$

$\therefore$ verification is in $O(V^2)$

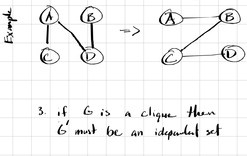
so problem is NP

Reduce from Clique

1. Define Clique, given $G = (V, E)$ and integer k, does G have a subgraph $S \subseteq V$ of size k in which all $v \in S$ are connected by a unique $e \in E$?
2. given a graph $G = (V, E)$ we can construct a graph where all $(u,v) \in E$ in G are not in G' and all $(u,v) \notin E$ are in G'

where $u \in V$ + $v \in V$

Example



3. if G is a clique then G' must be an independent set

4. the construction of G' from G is done by copying vertices $O(V)$ and inverting edges $O(E)$

$\therefore$ constructing the inverse Graph is linear time $\max\{O(V), O(E)\}$